

Case Study: Austech Manufacturing Alliance (AMA) - Implementation of an Enterprise Resource Planning System

Nature of the business

The Austech Manufacturing Alliance (AMA) is a network of nine mid-sized manufacturing plants spread across Victoria and South Australia. Each plant specialises in different product lines, including automotive parts, industrial machinery, agricultural equipment and specialised steel fabrication. The alliance was formed only 18 months ago to strengthen competitiveness against international imports, reduce operational inefficiencies and coordinate supply chains more effectively.

While united under the AMA umbrella, each plant still operates largely independently, with its own management team, IT setup and financial systems. The largest plant, Austech Geelong, employs over 600 staff and is the only site with a dedicated CIO and technology team. Smaller plants such as Whyalla Steel Components and Horsham Agri-Engineering rely on local IT contractors or tech-savvy staff members doubling up as informal “IT managers”.

All plants are significant energy users. Their operations rely heavily on electricity for machining, welding, compressed air, heat treatment, robotics and IT infrastructure. Energy costs are a major component of overall operating expenses, particularly in South Australia, where wholesale prices and volatility have historically been higher. The alliance leadership is increasingly aware that Australia’s electricity system is changing rapidly – with record power demand, rising renewable penetration and growing use of batteries – and that AMA’s future competitiveness will depend in part on how well it manages energy as well as information.

The alliance’s strategic intent is to move towards greater centralisation of finance, HR, supply chain and production systems to streamline operations, reduce duplication and gain a competitive edge. At the same time, it wants to improve visibility of energy usage and emissions across all plants and respond more intelligently to price signals and grid conditions in a national market that is undergoing a major transition.

The problems

Currently, the alliance plants use fragmented, outdated and inconsistent systems. This is affecting not only financial and operational performance but also AMA’s ability to respond to shifts in the energy market.

Finance and HR processes differ from plant to plant. Some sites use modern cloud-based accounting tools, while others still rely on spreadsheets and paper-based payroll. Supply chains are managed in silos, with each plant negotiating separate contracts and managing its own suppliers. Production scheduling and inventory management approaches range from advanced digital dashboards at Geelong to manual whiteboards and paper slips at Whyalla. Maintenance and asset tracking are similarly inconsistent, with equipment downtime poorly tracked and often underestimated, leading to costly surprises.

These disparities have started to strain productivity and erode margins. Customers increasingly demand faster delivery, tighter quality assurance and greater transparency into sourcing and environmental performance. AMA is struggling to provide consolidated, real-time information across plants and cannot easily demonstrate its energy or emissions profile to major clients.

At the same time, Australia’s electricity system is changing in ways that directly affect AMA:

- Demand for power on the National Electricity Market (NEM), which supplies Victoria and South Australia, has reached record highs in the final quarter of recent years, driven by greater heating and cooling needs, electrification, the rise of power-hungry data centres and population growth.

- Renewable energy (including rooftop solar, grid-scale solar, wind and batteries) has, for the first time, supplied more than half of the NEM's demand over a full quarter. Renewable generation now routinely displaces higher-cost coal and gas generation during many hours of the day, pushing average wholesale prices down.
- Rooftop solar alone has reached new records, supplying a significant share of demand and at times covering more than 60 per cent of total system needs. This leads to periods of very low daytime operational demand on the grid, forcing conventional generators to ramp down.
- The growth of utility-scale wind and solar, combined with an expanding fleet of large and small batteries, is reshaping when and how electricity is generated and priced. Battery discharge during peaks has helped reduce reliance on gas and hydro in those periods and contributed to lower average wholesale prices and fewer high-priced intervals.

For AMA, this evolving landscape presents both opportunities and risks. On one hand, increased renewable output has driven average wholesale prices lower, particularly in Victoria and South Australia, with quarter-average prices in some regions falling to around \$37–\$50 MWh. On the other hand, the system is experiencing more extremes:

- Periods of very low demand in the middle of the day due to rooftop solar, when prices can fall and renewable generation is sometimes curtailed or “economically offloaded” due to network constraints.
- Periods of very high demand during heatwaves or when renewable output drops, especially in South Australia, where spot prices can briefly surge to the market cap (e.g. \$20,300 MWh) as gas-fired generation and imports from other states are called upon.

AMA plants, especially those in South Australia, can face highly variable electricity costs depending on when they run their most energy-intensive processes. Yet, because each site operates with separate systems and limited data, the alliance does not have a consolidated view of its energy usage by plant, by product or by time of day. Production scheduling takes little account of electricity price patterns, and there is no central strategy to shift flexible loads into cheaper periods or to make use of low-cost renewable energy when it is abundant.

Some plants have installed rooftop solar panels or are considering on-site batteries to reduce exposure to grid prices and improve sustainability credentials. However, these initiatives are local, uncoordinated and not integrated into any central reporting or optimisation framework. AMA cannot easily quantify the benefits, monitor curtailment issues, or demonstrate the environmental benefits in customer reporting.

The AMA Board has approved a budget of \$3.2 million to explore the implementation of an Enterprise Resource Planning (ERP) system that would unify finance, HR, supply chain and production management across all nine plants. There is an emerging expectation that the ERP should also provide the foundation for better energy and sustainability reporting, enabling AMA to respond more effectively to the realities of a high-renewables, increasingly data-driven electricity system.

The people

Daniel Carter – CIO, Austech Geelong

Daniel is a visionary but pragmatic CIO. He leads the only dedicated technology team in the alliance. Daniel strongly advocates for a cloud-first ERP solution to modernise AMA, integrate processes and data across plants, and keep pace with global competitors. He sees the ERP as a critical platform for integrating not just finance and supply chain data but also production and energy consumption data, allowing AMA to understand where and when it uses power, how that aligns with renewables availability, and how to schedule energy-intensive activities more intelligently.

However, he knows his team of five cannot handle implementation alone and that he must overcome significant resistance and budget constraints.

Linda Huang – CFO, AMA Board

Linda is extremely cost-conscious and worried about overspending. She fears a drawn-out ERP implementation that could spiral beyond the \$3.2 million capital budget and is sceptical of software-as-a-service subscription models, preferring an on-premises solution with predictable capital costs. At the same time, she is acutely aware that energy costs are now a major line item and that volatility can impact margins, especially when spot prices spike. She wants better reporting on energy costs, the ability to compare plants' energy efficiency and a clearer picture of how renewable penetration and falling average wholesale prices can be translated into sustained cost savings for AMA. She is cautious about cloud ERP, partly because of perceived dependence on large data centres which themselves are contributing to power demand growth.

Greg O'Donnell – Plant Manager, Whyalla Steel Components

Greg is an old-school operator and strongly resistant to change. He fears that centralisation will mean "head office" interference and potential job losses for his loyal staff. He is sceptical of claims that an ERP can help with real problems on the factory floor. He also views discussions about energy optimisation and renewables with suspicion, seeing them as distractions from meeting production targets. His plant occasionally experiences price spikes due to South Australia's volatile market, but he considers this "just part of doing business" and has not systematically explored how scheduling or efficiency measures could mitigate exposure.

Sofia Mendes – Operations Director, Horsham Agri-Engineering

Sofia is pragmatic and supportive of efficiency improvements but is concerned about disruption to production schedules during the ERP rollout. She has already lost two large contracts due to late deliveries and sees better production planning and supply chain visibility as urgent needs. She is also interested in how an ERP could help align production runs with times when electricity is cheaper and more renewable energy is available, particularly given that her plant has recently installed rooftop solar but lacks good tools to track its generation and usage.

Arjun Patel – IT Contractor, Mildura Plant

Arjun is a long-standing external IT contractor who supports the Mildura plant. He is defensive about his role, as centralisation and a unified ERP could make his services redundant. He raises concerns about cybersecurity and reliability in a cloud-based system, particularly in a power market that is experiencing more extremes and occasional intervention by AEMO to maintain system strength. Arjun questions whether AMA has fully considered risks such as grid outages, data centre dependencies and complexity introduced by integrating energy and production data.

Tanya Singh – Union Representative

Tanya represents frontline factory workers across the alliance. She is deeply suspicious of the ERP project and worries it will be used to justify "efficiency-driven downsizing". She also hears members' concerns about rising living costs and job security in a changing economy. While energy issues are not her primary focus, she has begun to notice that some plants are experimenting with changing shift patterns or overtime approvals in response to "energy cost pressures", and she wants to ensure that any new scheduling approaches driven by the ERP consider worker wellbeing and fair consultation.

Current Systems

The systems landscape is fragmented and inconsistent across the plants, with little central control or transparency.

Finance and HR

- Geelong uses SAP Business One, but with outdated patches and limited integration with other systems.
- Whyalla relies on Excel spreadsheets and MYOB desktop software with no integration, making consolidation slow and error-prone.
- Horsham uses Xero Cloud for finance, but HR remains largely manual.
- Payroll cycles are inconsistent across plants, leading to delays and reconciliation issues.

There is no single source of truth for financial performance or staff information, and no standard way to allocate or analyse energy costs across plants or product lines.

Supply Chain and Procurement

Suppliers are managed separately at each plant, often with duplicated suppliers under different terms. There is no consolidated procurement system to leverage bulk discounts or to manage contracts consistently. Manual purchase order approvals introduce bottlenecks and increase the risk of errors. AMA has no central visibility of how energy and other input costs flow through the supply chain or how they affect profitability across products.

Production and Inventory

Production scheduling and inventory management vary significantly:

- Geelong uses digital dashboards to plan and monitor production, but these tools are bespoke and not linked to other plants.
- Whyalla still uses whiteboards and paper slips, which are prone to error and make it difficult to adjust plans quickly in response to demand changes or equipment issues.
- Mildura relies on a bespoke Microsoft Access database that is no longer supported and at risk of failure.

Inventory mismatches are common, creating “phantom stock” issues where records do not match reality. Lead times are unreliable, contributing to late deliveries and customer dissatisfaction.

Production planning currently takes little account of energy considerations. Most plants run energy-intensive processes during standard hours regardless of price or grid conditions. There is no central system to help identify which processes could be shifted to low-price periods, such as when wind and solar output is high, or to ensure that plants are not inadvertently concentrating peak loads during high-price, low-renewables intervals.

Request for Proposal

The AMA Board has issued a Request for Proposal (RFP) for an implementation partner to deliver a unified ERP system across all nine plants. The consultancy is expected to address both the traditional ERP scope and the emerging challenges and opportunities associated with Australia’s energy transition.

The consultancy will be expected to:

1. Conduct an initial discovery exercise to assess current systems, data and organisational readiness across all plants, including an assessment of how energy is currently measured, reported and managed.
2. Analyse cloud versus on-premises ERP options, including risks, costs and benefits, with explicit consideration of reliability and security in a power system characterised by growing renewable penetration, increased use of batteries and evolving demand patterns.
3. Recommend and support vendor selection (e.g. SAP, Oracle, Microsoft Dynamics), considering not only core ERP capabilities but also the ability to integrate with energy monitoring systems, support sustainability and emissions reporting, and scale with AMA's future needs.
4. Develop an implementation roadmap that unifies finance, HR, supply chain, production, maintenance and, where feasible, energy data. The roadmap should include clear phases, prioritising quick wins (such as basic standardisation and reporting) while managing risk in more complex integration areas.
5. Design risk mitigation and change management strategies that address:
 - Resistance from plant managers and staff.
 - Concerns about job security and centralisation.
 - Potential disruption to production schedules.
 - Dependencies on external energy and IT infrastructure.
6. Provide training programmes tailored to management, factory staff and support functions, focusing on how to use the ERP for day-to-day operations, as well as how to interpret and act on integrated financial, operational and energy data.
7. Manage rollout, testing and support across all alliance members, ensuring that lessons from early sites are captured and applied to subsequent phases.

The Assignment

Your consulting organisation has decided to respond to the RFP. As the Director on your team, it has fallen to you to lead the response.

You will need to prepare the response in consideration AMA's request – i.e. to cover what their business problem is and how you will execute the implementation of an ERP. The structure of the proposal should follow and incorporate the standard flow that your consultancy's proposals typically follow:

- Executive summary
- Introduction
- Our understand of your needs
- Engagement methodology and timelines (including risk management and governance)
- Pricing and assumptions
- Team CVs

- Similar work we have performed